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Sheet (2)

#2.5 سلمناها للدكتور
#2.6 ومش بحاية فاينل

#2.1 $X(k)=\sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk}$

$$X(k+N)=\sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi n(k+N)}{N}}$$

$$=\sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk}\; e^{-j2\pi n}$$

$e^{-j2\pi n}=1$

$$=\sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi}{N}nk} = X(k)$$

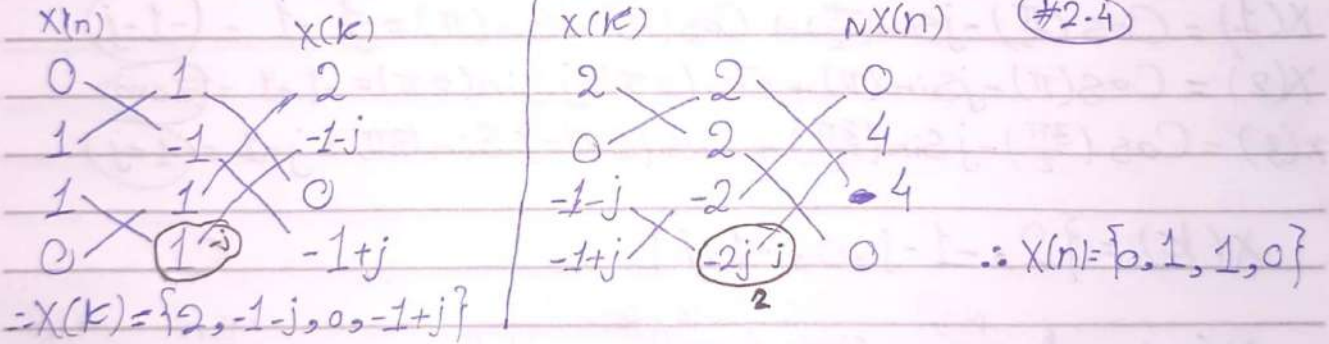
$\therefore X(k)=X(k+N)$

↳ Sequence of $X(k)$ is Periodic every N Samples.

#2.2 ~~DFT~~ FFT ? IFFT ?

$X(n)=\{0,1,1,0\}$ $(N=4)$

لحل السؤال #2.4



$\therefore X(k)=\{2,-1-j,0,-1+j\}$

$\therefore x(n)=\{0,1,1,0\}$

#2.3 Parseval's Theorem

$$\sum_{n=0}^{N-1} x^2(n)=\frac{1}{N}\sum_{k=0}^{N-1}|X(k)|^2$$

$|X(k)|=\sqrt{Re^2+Img^2}$

↳ $\{2,\sqrt{2},0,\sqrt{2}\}$

L.H.S $=(0^2+1^2+1^2+0^2)=2$

R.H.S $=\frac{1}{4}\left(2^2+(\sqrt{2})^2+0^2+(\sqrt{2})^2\right)=\frac{8}{4}=2$

$\therefore$ Parseval's Theorem is verified.

	N. of Add	N. of Mul
DFT	$4\times3=12$	$4^2=16$
FFT	$4\log_2(4)=8$	$\frac{4}{2}\log_2(4)=4$
Saving	4	12

#2.4 $x(n)=\{0,1,1,0\}$ DFT?

$$X(k)=\sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi}{N}nk}$$

$$=\underbrace{x(0)}_{\text{zero}}+\underbrace{x(1)}_{1}e^{-j\frac{\pi}{2}k}+\underbrace{x(2)}_{1}e^{-j\pi k}+\underbrace{x(3)}_{\text{Zero}}e^{-j\frac{3\pi}{2}k}$$

$$X(k)=e^{-j\frac{\pi}{2}k}+e^{-j\pi k}$$

$X(0)=1+1=2$

$X(1)=\cos(\frac{\pi}{2})-j\sin(\frac{\pi}{2})+\cos(\pi)-j\sin(\pi)=-j-1=-1-j$

$X(2)=\cos(\pi)-j\sin(\pi)+\cos(2\pi)-j\sin(2\pi)=-1+1=\text{Zero}$

$x(3)=\cos(\frac{3\pi}{2})-j\sin(\frac{3\pi}{2})+\cos(3\pi)-j\sin(3\pi)=j-1=-1+j$

$X(k)=\{2,-1-j,0,-1+j\}$

$$x(n)=\frac{1}{N}\sum_{k=0}^{N-1}X(k)e^{j\frac{2\pi}{N}nk}=\frac{1}{4}\left[X(0)+X(1)e^{j\frac{\pi}{2}n}+X(2)e^{j\pi n}+X(3)e^{j\frac{3\pi}{2}n}\right]$$

$$x(n)=\frac{1}{4}\left[2+(-1-j)e^{j\frac{\pi}{2}n}+(-1+j)e^{j\frac{3\pi}{2}n}\right]$$

$$x(1)=\frac{1}{4}\left[2+(-1-j)\left(\cos(\tfrac{\pi}{2})+j\sin(\tfrac{\pi}{2})\right)+(-1+j)\left(\cos(\tfrac{3\pi}{2})+j\sin(\tfrac{3\pi}{2})\right)\right]$$

$$=\frac{1}{4}[2+(-j+1)+(j+1)]=1$$

$$x(0)=\frac{1}{4}[2+(-1-j)+(-1+j)]=0$$

$$x(2)=\frac{1}{4}\left[2+(-1-j)\left(\cos(\pi)+j\sin(\pi)\right)+(-1+j)\left(\cos(3\pi)+j\sin(3\pi)\right)\right]$$

$$=\frac{1}{4}[2+1+j+1-j]=1$$

$$x(3)=\frac{1}{4}\left[2+(-1-j)\left(\cos(\tfrac{3\pi}{2})+j\sin(\tfrac{3\pi}{2})\right)+(-1+j)\left(\cos(\tfrac{2\pi}{2})+j\sin(\tfrac{2\pi}{2})\right)\right]$$

$$=\frac{1}{4}[2+j-1-j-1]=0 \qquad x(n)=\{0,1,1,0\}$$